The Dynamics of Internal Migration: A New Fact and its Implications

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We need good models of internal migration

- Internal migration...
  - ...is a key method of adjustment to local economic shocks
    - Labor demand (Blanchard and Katz, 1992)
    - Trade (Caliendo, Dvorkin and Parro, 2019)
    - Industrialization (Tombe and Zhu, 2019)
    - Technological change (Giannone, 2017; Rubinton, 2020; Faber, Sarto and Tabellini, 2022; Kleinman, Liu and Redding, 2023; Davis, Fisher and Veracierto, 2021)
    - Taxes (Kleven, Landais, Muñoz and Stantcheva, 2020)
    - Global warming (Cruz and Rossi-Hansberg, 2021)
  - ...provides insurance (Giannone, Li, Paixao and Pang, 2020)
  - ...amplifies local economic disparities (Howard, 2020)
  - ...drives sorting by class and race (Shertzer and Walsh, 2019; Diamond, 2016)
  - ...is an important determinant of housing cycles (Howard and Liebersohn, 2021, 2023)

- Old literature has used static models of location choice (Rosen, 1979; Roback, 1982)
  - Models net migration not gross migration
- New literature has built dynamic models and explicitly modeled migration
- Two major facts motivate these new models:
  1. Internal migration is rare
  2. People move to nearby and populous places (gravity)

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(a) Moving costs?

(b) Persistence in unobserved match-specific characteristics?

Does it matter how we understand these two facts?

- New literature has emphasized moving costs
  - Tractable
  - Easily matches both facts
  - Natural extension of the trade literature

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2. New model
   - Idiosynchratic unobservables correlated across space and time
   - Model can match new fact, while also matching existing migration patterns
3. New Implications of the model
   - Short-run elasticities are still similar to existing models
   - Long-run elasticities are very different
   - Speed of population adjustment is very different
   - Revisit classic spatial question: which places are getting better to live?

New Fact about Internal Migration

Data

- Gies Consumer and Small Business Credit Panel (GCCP)
  - Credit data from one of the leading providers of credit reports
  - 1 percent of Americans with credit reports
  - Includes state of residence
  - Panel data, 2004-2018
  - Migration is rare
  - Migration follows a gravity pattern
- For robustness, also use the PSID

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New Fact

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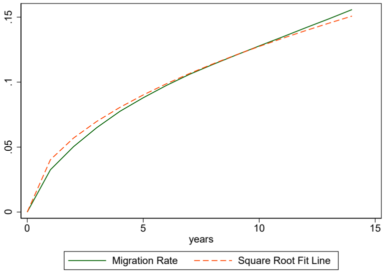
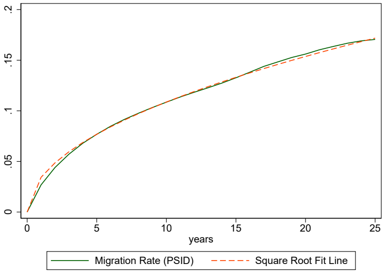


(a) GCCP



(b) PSID

Standard Moving Cost Model

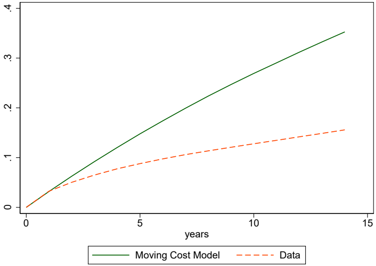
I locations indexed by i, N individuals indexed by n, and discrete time indexed by t:

- Agents choose location that maximizes utility

$$V_{nt}(i) = \max_{j}\{v_{jt} - \delta_{ij} + \epsilon_{jnt} + \mathbb{E}[V_{nt+1}(j)]\}$$

- ϵ_{jnt} is i.i.d. and has a Gumbel distribution

Dynamics of Migration



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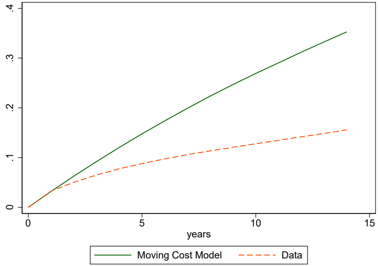

- In standard model, migration is Markov
  - When migration is rare, t-year migration proportional to t, not $\sqrt{t}$
- Addition of simple state variables does not work
  - Age
  - Last year's location
  - Birth state
- Can be reconciled with flexible tenure-dependent moving costs (i.e. many fine-tuned parameters)

The SPACE Model

Model

I locations indexed by i, continuum of individuals indexed by n, and discrete time indexed by t:

- Agents choose location that maximizes utility

$$V_{nt}(\epsilon_{nt}) = \max_j \{v_{jt} + \epsilon_{jnt}\} + \beta \mathbb{E}[V_{nt+1}(\epsilon_{nt+1}) | \vec{\epsilon}_{nt}]$$

- No moving costs
- State variable is match-specific idiosyncratic unobservables for every location
- ϵ will be correlated across j and t
    - Spatially and Persistently Auto-Correlated Epsilons (SPACE)

SPACE Model: Spatial Correlation

- Agents choose locations to maximize utility:

$$V_{nt}(\epsilon_{nt}) = \max_i \{v_i + \epsilon_{int}\} + \beta \mathbb{E} V_{nt+1}(\epsilon_{nt+1})$$

- ϵ_{nt} has a Generalized Extreme Value (GEV) distribution:

$$F(\epsilon_{1nt}, \ldots, \epsilon_{Int}) = \exp\left(-G(e^{-\epsilon_{1nt}}, \ldots, e^{-\epsilon_{Int}})\right)$$

- where G is: (McFadden, 1977)
  - Non-negative
  - Homogeneous of degree 1.
  - Cross-partials of k elements are nonnegative if k is odd, nonpositive if k is even.
  - $G \to \infty$ when any argument $\to \infty$.
- Marginal of ϵ_{int} is a standard Gumbel distribution
- Allows for correlated choices across locations

Populations

- Probability of agent n choosing location i:

$$p_i = e^{v_i} \frac{G_i(e^{v_1}, \ldots, e^{v_I})}{G(e^{v_1}, \ldots, e^{v_I})}$$

  - G_i is the partial derivative of G with respect to i:

$$G_i(x_1, \ldots, x_I) = \frac{\partial G(x_1, \ldots, x_I)}{\partial x_i}$$

- Examples of GEV include simple logit, nested logit, and cross-nested logit

Migration: Space and Time Correlation

- To introduce correlation over time, we extend the GEV model:

$$(\epsilon_{nt}, \epsilon_{nt+1}) \sim F_2(\cdot, \cdot)$$

- Joint distribution of utilities across 2 time periods:

$$\begin{aligned} &F_2(\epsilon_{1nt}, \dots, \epsilon_{Int}, \epsilon_{1nt+1}, \dots, \epsilon_{Int+1}) \\ &\quad = \exp\left(-G(H(e^{-\epsilon_{1nt}}, e^{-\epsilon_{1nt+1}}), \dots, H(e^{-\epsilon_{Int}}, e^{-\epsilon_{Int+1}}))\right) \end{aligned}$$

- $H(x_1, x_2) = \left(x_1^{\frac{1}{1-\rho}} + x_2^{\frac{1}{1-\rho}}\right)^{1-\rho}$, where ρ governs correlation across time.
- Within a period, distribution of ϵ_{nt} is the same as before
- Joint distribution implies a conditional distribution of ϵ_{nt+1} given ϵ_{nt}
- ϵ_{nt} evolves in a Markov process where each ϵ_{nt+1} is distributed according to the conditional distribution of ϵ_{nt}

Closed-Form Solution for Steady-State Migration Rates

- In steady-state, the migration rate from location i to location j is given by a closed-form solution:

$$m_{i \to j} = (1 - \rho) e^{v_i + v_j} \frac{G_i G_j - G_{ij} G}{G^2}$$

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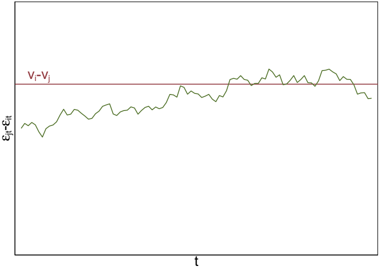
- Rewrite as a gravity equation:

$$m_{i\to j} = (1-\rho)p_ip_j\left(1+\tau_{ij}\right)$$

  - Where $\tau_{ij} = \frac{-G_{ij}G}{G_iG_j}$, loosely representing the correlation between choices i and j.

Dynamics of Migration

- Recall: t-year migration rate is approximately $\propto \sqrt{t}$
- How does the SPACE model match this fact?



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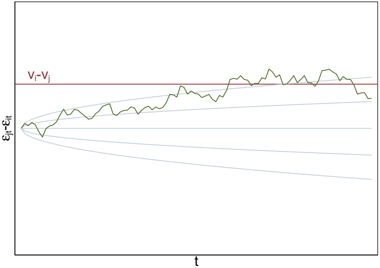

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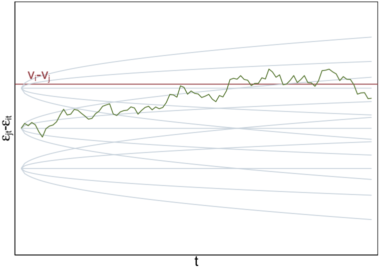

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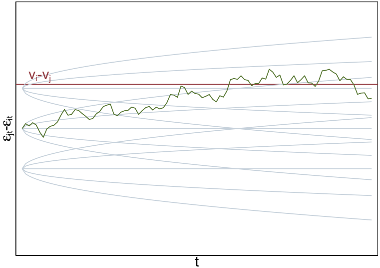

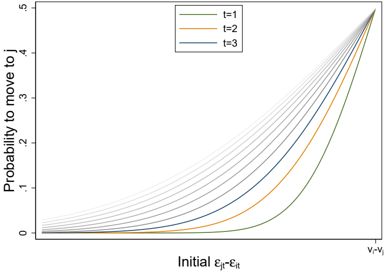

$$m_{i\to j,t}\big|\epsilon_{j0}-\epsilon_{i0} \approx \Phi\left(\frac{\epsilon_{j0}-\epsilon_{i0}-(v_i-v_j)}{\sigma\sqrt{t}}\right)$$

Square Root Fact

Proposition

In the SPACE model,

$$1 \leq \lim_{\rho \to 1} \frac{m_{i \to j,t}}{m_{i \to j,1}} \frac{1}{\sqrt{t}} \leq 1.06$$

Model Calibration

- In general, individual's dynamic location choices hard to simulate—requires drawing from GEVs
  - In many applications, not an issue

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- In general, individual's dynamic location choices hard to simulate—requires drawing from GEVs
  - In many applications, not an issue
- I propose a specific calibration:

$$G(x_1, \ldots, x_I) = \sum_i w_i x_i + \sum_{i \neq j} \left((w_{ij} x_i)^{\frac{1}{1-\gamma_{ij}}} + (w_{ji} x_j)^{\frac{1}{1-\gamma_{ij}}} \right)^{1-\gamma_{ij}}$$

  - $\gamma_{ij} < \rho$
  - $w_i = p_i p_i e^{-v_i}, \quad w_{ij} = p_i p_j e^{-v_i} 2^{\gamma_{ij}}$
  - Cross-nested logit
  - Initial populations match exactly
  - $m_{i \to j} = \frac{1-\rho}{1-\gamma_{ij}} p_i p_j$

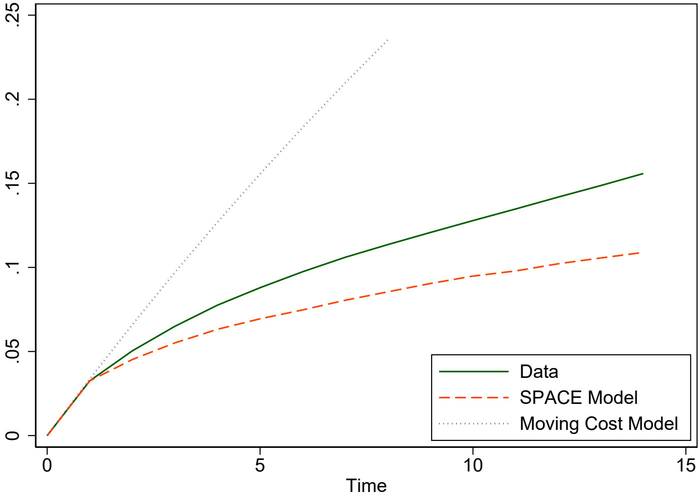
Model Calibration Continued

- Assume $\rho \to 1$ and $\gamma_{ij} \to 1$, but

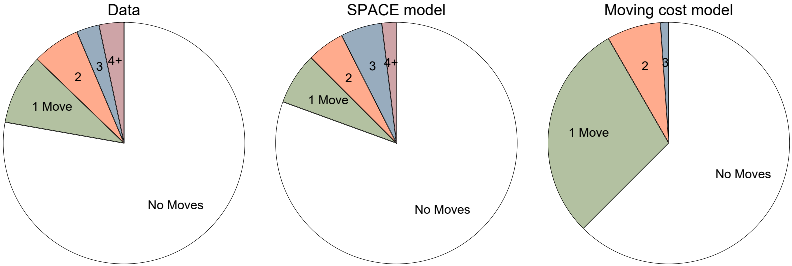
$$\frac{1-\rho}{1-\gamma_{ij}} \to 1 - \tilde{\rho}_{ij}$$

- Agents never move between nests, but within nests, they will sometimes migrate at a rate $m_{i \to j} = (1 - \tilde{\rho}_{ij}) p_i p_j$
- Reasonable to simulate:
  - Shares in each nest of the logit are fixed
  - Relative utility within a nest maps to relative utility of two independent Gumbels that have persistence governed by $\tilde{\rho}_{ij}$.

SPACE dynamics

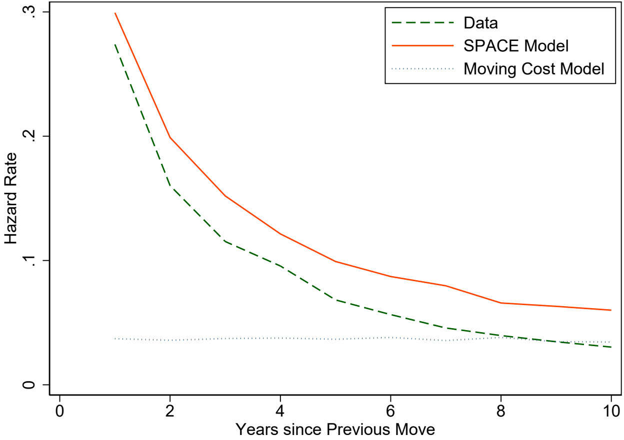


Frequency of Migration



Hazard Rate of Migration

- Conditional probability of moving after previous move



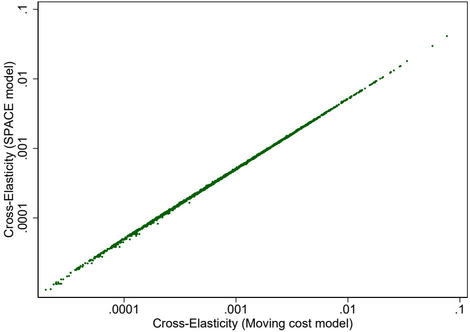
Implications of the Model

vs.

Short-run Population Cross-elasticity

- In the short-run, elasticities from moving cost model and SPACE model are quite similar

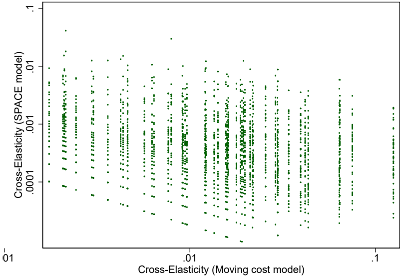
$$\underbrace{\frac{\partial p_i}{\partial v_j} \propto m_{i\to j}}_{\text{SPACE Model}} \qquad \underbrace{\frac{\partial p_i}{\partial v_j}_{\text{Moving Cost}} \propto \sum_k m_{i\to k} m_{k\to j}}_{\text{Moving Cost Model}}$$



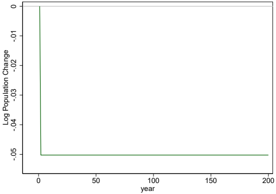
Long-run Population Cross-elasticity

- SPACE model has same cross-elasticities as short-run
- Moving cost model converges to approximately static logit

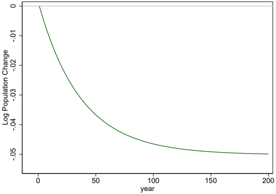
$$\underbrace{\frac{\partial p_i}{\partial v_j} \propto m_{i \to j}}_{\text{SPACE Model}} \qquad \underbrace{\frac{\partial p_i}{\partial v_j} \propto p_j}_{\text{Moving Cost Model}} \quad \text{(approximately)}$$



Population dynamics after a one-time permanent change in $v_{\text{Louisiana}}$



(a) Louisiana, SPACE model



(b) Louisiana, Moving cost model

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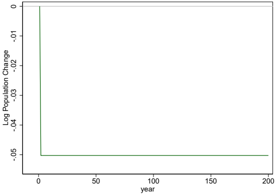

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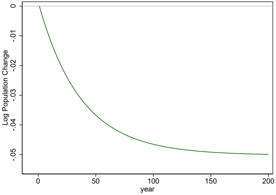

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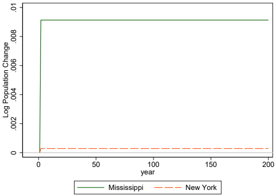
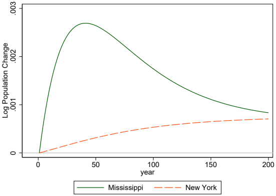


(c) Mississippi and New York, SPACE model



(d) Mississippi and New York, Moving cost model

Which states became better places to live?

- Use the SPACE model and observed populations to back out implied changes in v_i
- Use the Moving cost model and observed migration to back out implied changes in v_i
- Compare 1990 to 2018

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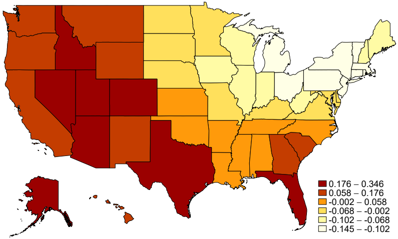
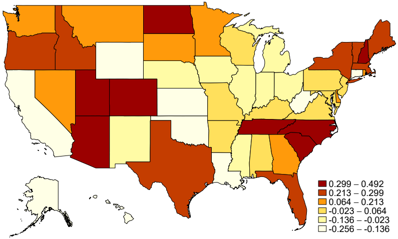


(a) SPACE Model



(b) Moving Cost Model

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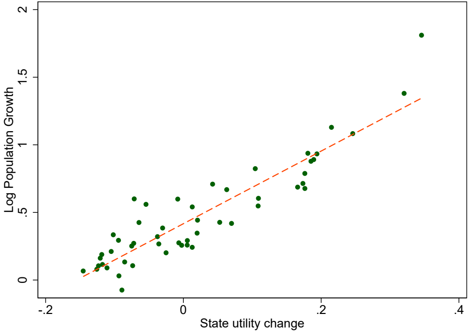

- Utility changes correlated but not identical to population growth
- Neighbor's utility changes matter, too

Takeaways

1. New fact: t-year internal migration rates are proportional to $\sqrt{t}$
2. SPACE model can match dynamic moments of migration
3. SPACE model has different implications for long-run adjustments, population dynamics, and moving costs

Migration is Rare

Comparison of interstate migration rates in IRS and GCCP

Return

Migration Follows a Gravity Pattern

Poisson regression:

$$\log m_{i\to j} = \beta \log \text{distance}_{ij} + \alpha \log p_i + \gamma \log p_j + \epsilon_{ij}$$

	(1) Migration (IRS)	(2) Migration (Credit)
Log Distance	-0.736***	-0.744***
	(0.0572)	(0.0515)
Log Origin Population	0.900***	0.923***
	(0.0832)	(0.0797)
Log Destination Population	0.822***	0.893***
	(0.0976)	(0.0799)
Observations	2550	2550

Standard Errors are two-way clustered by origin and destination states

* $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

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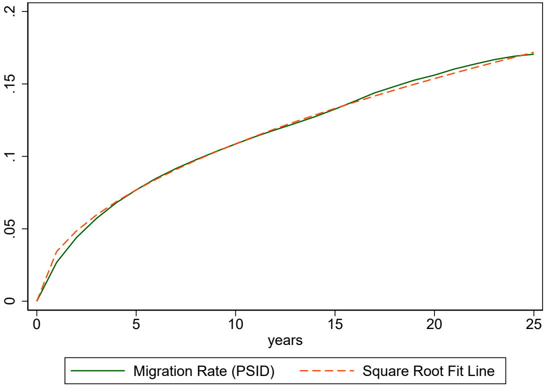
Square Root Rule, PSID



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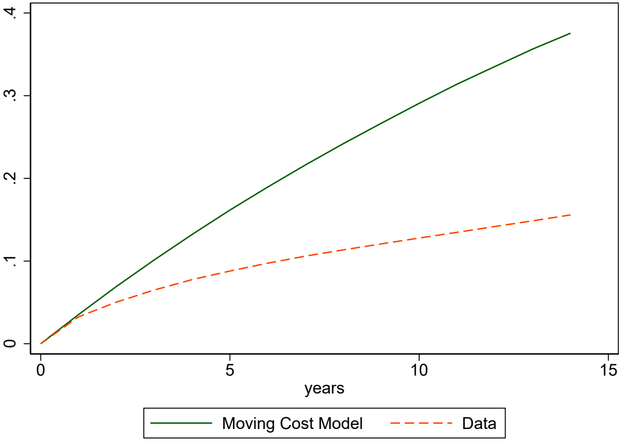
Adding Age as a State Variable Does Not Match the Square Root Pattern



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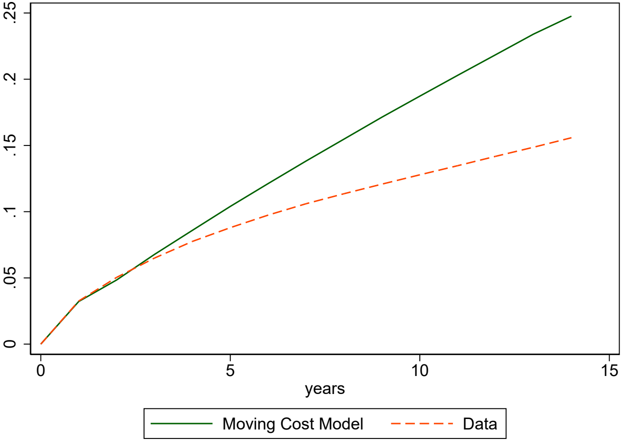
Adding Previous Location as a State Variable Does Not Match the Square Root Pattern



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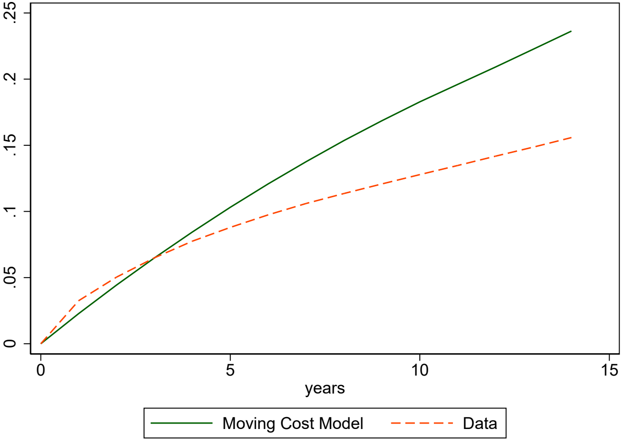
Adding Birthplace as a State Variable Does Not Match the Square Root Pattern



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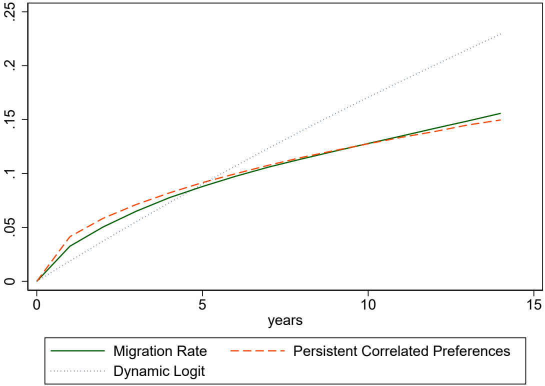
Square Root Rule, 5-year calibration

Proposition 2

As $\rho \to 1$, the t-year migration rate is proportional to $\sqrt{t}$.



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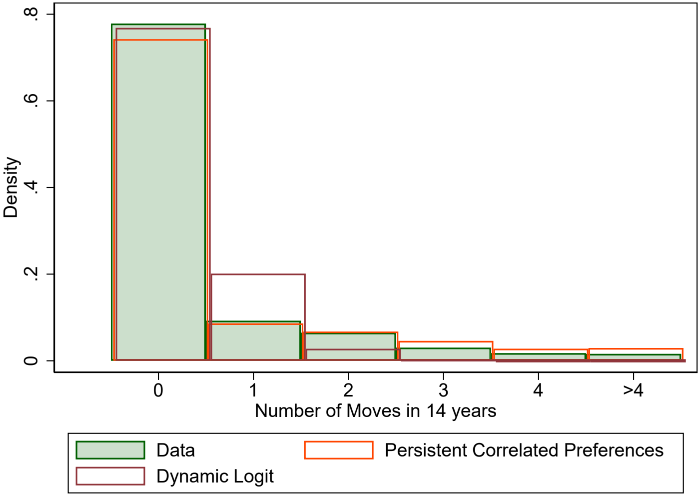
Frequency of Migration, 5-year calibration



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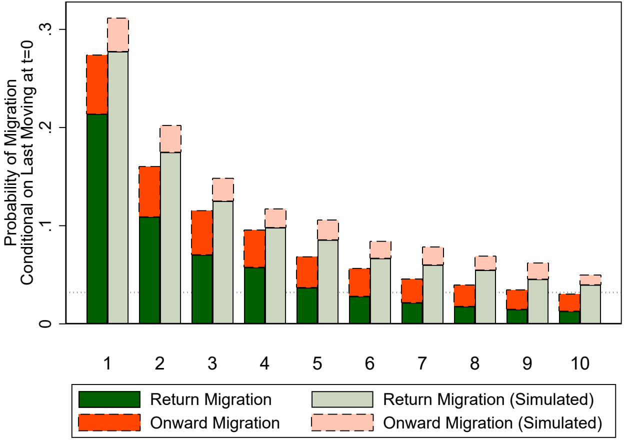
Return Migration, 5-year calibration

- Conditional probability of moving after previous move



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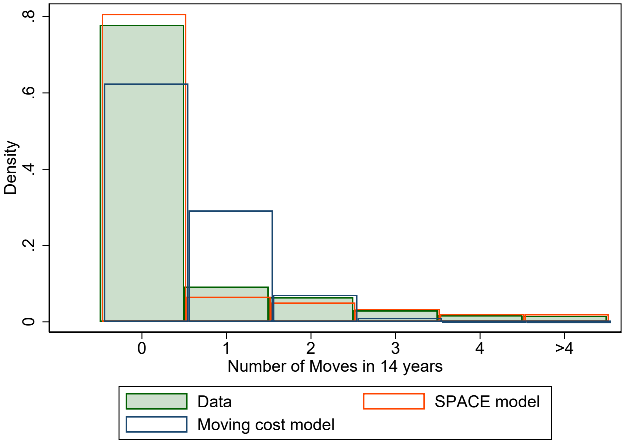
Frequency of Moves, Dynamic Logit Model



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Return Migration, Dynamic Logit Model

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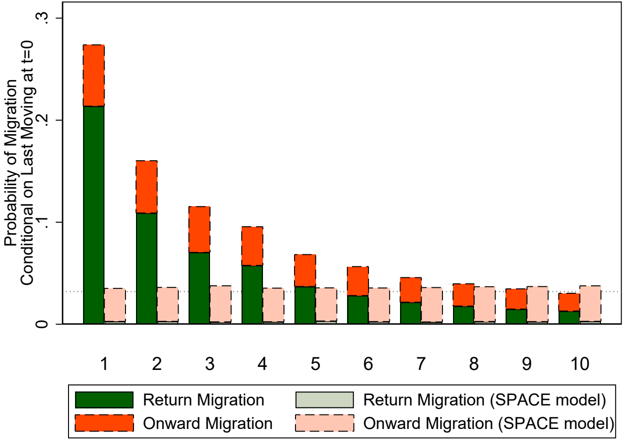

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